Truncated Microlocal Morse Theory

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1 Introduction

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Microlocal sheaf theory allows us to study the propagation of cohomology of complexes of sheaves. A central notion in the theory is that of microsupport of a complex of sheaves. It was defined by Kashiwara and Schapira in 1982 [KS82].

Let M be a real manifold and F a complex of sheaves. Then the microsupport $\mathrm{SS}(F)$ of F is a closed conic (i.e., stable under $\mathbf{R}^+$-action) coisotropic subset of the cotangent bundle of M. Roughly speaking, $\mathrm{SS}(F)$ describes the (co)directions of M in which the cohomology of some degree of F cannot be extended. Note that we cannnot regain from $\mathrm{SS}(F)$ information of the cohomological degree of F where its cohomology does not propagate.

However, there are certain propagation phenomena which depend on the degree of cohomology. One cannot treat such phenomena by microsupports. To address this difficulty, Kashiwara, Monteiro-Fernandes, and Schapira [KMS03a] introduced the notion of truncated microsupport. This notion encodes information about singularities in cotangent directions depending on the degree of cohomology.

Both notions above encode microlocal information. Then how can we deduce the global propagation results from this microlocal information? For ordinary microsupport, an answer is given by the non-characteristic deformation lemma due to Kashiwara [Kas75]. This lemma plays a fundamental role in microlocal sheaf theory, and is used throughout [KS90].

In this paper, we prove a truncated version of the non-characteristic deformation lemma with slight modifications. More precisely, we make two modifications to the lemma.

First, we weaken the hypothesis of vanishing of cohomologies. In the original lemma, we assume that the cohomologies of all degrees vanish. In our version, it is required only in degrees at most a fixed integer.

Second, we make an additional assumption on the deformation family. Note that we cannot apply the original proof of the lemma to our truncated version. Indeed we give an example showing that the conclusion of the truncated version can fail without this additional assumption.

To make the last assumption acceptable to the reader, we give several examples of deformation families with figures.

As an application, we prove a truncated version of the microlocal Morse lemma for a morphism of manifolds. By considering the sheaf of hyperfunction solutions of a $\mathcal{D}$-module, one can obtain a global version of Holmgren's uniqueness theorem.

Analytic point of view From the viewpoint of propagation, our results are along the lines of the Cauchy-Kowalevsky theorem. In 1971, Zerner showed a theorem on analytic extension of solutions of a single differential equation whose coefficients are holomorphic [Zer71]. In 1972, Bony and Schapira generalized the result to the case of families of differential operators [BS72]. In 1975, Kashiwara used this method and the Mittag-Leffler argument for sheaves to extend the result to the case of the solution complex of a coherent $\mathcal{D}$-module [Kas75]. Note that Kashiwara used this method to prove the constructibility theorem: if $\mathcal{M}$ is a holonomic $\mathcal{D}$-module, the solution complex $\mathrm{R}\mathscr{H}om_{\mathcal{D}_X}(\mathcal{M}, \mathcal{O}_X)$ is constructible.

Our result in the case of hyperfunction solutions can be seen as a truncated version of these propagation results.

Organization The paper is organized as follows. In Section 2, we review notation and results from [KS90]. In Section 3, we recall the definition of truncated microsupport, introduced in [KMS03a]. In Section 4, we prove a truncated version of the noncharacteristic deformation lemma. In the proof, Lemma 4.4 plays a crucial role. In Section 5, we apply the result in Section 4 to microlocal Morse theory. Finally in Appendix A, we give some technical results.

2 Notations and Review

In this section, we recall some definitions and results of homological algebra and sheaf theory including the theory of derived categories from [KS90], following its notations with slight modifications.

Let X be a topological space, and Z a locally closed set of X. Let us denote by $i\colon Z \hookrightarrow X$ the inclusion. We use the following functors:

$$\mathrm{R}\Gamma_Z(\,\cdot\,) \coloneqq \mathrm{R}i_* i^!, \quad (\,\cdot\,)_Z \coloneqq \mathrm{R}i_! i^{-1}.$$

If there is no risk of confusion, we will write $\mathbf{k}_Z$ rather than $\mathbf{k}_{X,Z}$.

2.1 Geometrical Notions

2.1.1 Proper Morphisms

We use the following proposition in the proof of Proposition 5.1

Proposition 2.1 (see [Wed16, Proposition 1.32])**.** Let X, Y, and Z be locally compact (Hausdorff, by definition) spaces, and let f, g be continuous. If $g \circ f$ is proper and f surjective, then g is proper.

2.1.2 The Fundamental Diagram of Cotangent Bundles

Let M be a real manifold of class C^∞ which we assume countable at infinity, i.e., a countable union of compact sets. We denote by $\tau_M\colon TM \to M$ the tangent bundle of M, and by $\pi_M\colon T^*M \to M$ the cotangent bundle of M. If there is no risk of confusion, we will write π rather than π_M. If $f\colon M \to N$ is a morphism of manifolds (i.e., a C^∞-map), then we have the following natural diagrams.

$$\begin{array}{ccccc} TM & \xrightarrow{f'} & M \times_N TN & \xrightarrow{f_\tau} & TN \\ & \searrow^{\tau_M} & \downarrow & \square & \downarrow{\scriptstyle \tau_N} \\ & & M & \xrightarrow[f]{} & N, \end{array} \qquad \begin{array}{ccccc} T^*M & \xleftarrow{f_d} & M \times_N T^*N & \xrightarrow{f_\pi} & T^*N \\ & \searrow^{\pi_M} & \downarrow & \square & \downarrow{\scriptstyle \pi_N} \\ & & M & \xrightarrow[f]{} & N. \end{array}$$

In these diagrams, f_τ and f_π denote the natural projections, f' is defined by $(x;v) \mapsto (x;df_x v)$, and f_d is induced by the transposition of f'.

2.2 Approximation by Inductive Limits

Proposition 2.2 ([KS90, Proposition 2.5.1])**.** Let X be a topological space, Z a subspace, and F a sheaf on X. Consider the canonical morphism:

$$\psi\colon \varinjlim_{U} \Gamma(U;F) \to \Gamma(Z;F),$$

where U ranges through the family of open neighborhoods of Z in X. Then the following hold.

(i) The morphism ψ is injective.

(ii) Assume that X is Hausdorff and Z is compact. Then ψ is an isomorphism.

(iii) Assume that X is paracompact and Z is closed. Then ψ is an isomorphism.

Note that a paracompact space is, by definition, a Hausdorff space.

2.3 Complement of Homological Algebra

Consider a projective system of abelian groups $(M_n, \rho_{n,p})_n$ indexed by $\mathbf{N}$. One says that this system satisfies the Mittag-Leffler condition (ML condition for short) if for any $n \in \mathbf{N}$, the decreasing sequence $(\rho_{n,p}(M_p))_p$ of subgroups of M_n is stationary. For a projective system of abelian groups $(M_n)_n$, we set

$$M_\infty \coloneqq \varprojlim_{n} M_n.$$

Consider a projective system of complexes

$$E_n^\bullet\colon \cdots \to M_n^{j-1} \to M_n^j \to M_n^{j+1} \to \cdots, \tag{2.1}$$

and its projective limit

$$E_\infty^\bullet\colon \cdots \to M_\infty^{j-1} \to M_\infty^j \to M_\infty^{j+1} \to \cdots. \tag{2.2}$$

Denote by

$$\Phi_k\colon H^k(E_\infty^\bullet) \to \varprojlim_{n} H^k(E_n^\bullet) \tag{2.3}$$

the natural morphism.

Proposition 2.3 ([KS90, Proposition 1.12.4])**.** Assume that for each $k \in \mathbf{Z}$, the system $\left(M_n^k\right)_n$ satisfies the ML condition. Then

(a) for each $k \in \mathbf{Z}$, the map Φ_k is surjective,

(b) if moreover, for a given i the system $\left(H^{i-1}(E_n^\bullet)\right)_n$ satisfies the ML condition, then Φ_i is bijective.

The following is called Kashiwara's lemma.

Proposition 2.4 ([KS90, Proposition 1.12.6])**.** Let $(X_s, \rho_{s,t})$ be a projective system of sets indexed by $\mathbf{R}$. Assume that for each $s \in \mathbf{R}$, the canonical maps

$$\lambda_s \colon X_s \to \varprojlim_{r<s} X_r$$

and

$$\mu_s \colon \varinjlim_{t>s} X_t \to X_s$$

are both injective (resp. surjective). Then all the maps ρ_{s_0,s_1} $(s_0 \leqq s_1)$ are injective (resp. surjective).

2.4 Mittag-Leffler Procedure for Sheaves

We use the following in the proof of Theorem 4.2.

Proposition 2.5 ([KS90, Proposition 2.7.1])**.** Let X be a topological space and let $F \in \mathsf{D}^+(\mathbf{k}_X)$. Let $(U_n)_{n\in\mathbf{N}}$ be an increasing sequence of open subsets of X, $(Z_n)_{n\in\mathbf{N}}$ a decreasing sequence of closed subsets of X. Set $U \coloneqq \bigcup_n U_n$, $Z \coloneqq \bigcap_n Z_n$. Then the following hold.

(i) For any j, the natural map $\varphi_j \colon H^j_Z(U;F) \to \varprojlim_n H^j_{Z_n}(U_n;F)$ is surjective.

(ii) Assume that for a given j, the projective system $(H^{j-1}_{Z_n}(U_n;F))_n$ satisfies the ML condition. Then φ_j is bijective.

(iii) Let now $(X_n)_{n\in\mathbf{N}}$ be an increasing family of subsets of X satisfying $X = \bigcup_n X_n$ and $X_n \subset \mathrm{Int}(X_{n+1})$ for all n. Assume that for a given j, the projective system $\left(H^{j-1}(X_n;F)\right)_n$ satisfies the ML condition. Then the natural map $H^j(X;F) \to \varprojlim_n H^j(X_n;F)$ is bijective.

3 Truncated Microsupport

In this section, we review the definition of a truncated microsupport and its functorial property under direct images of proper morphisms. First, we review the original definition of a microsupport.

Definition 3.1 ([KS90, Definition 5.1.2])**.** Let $F \in \mathsf{D}^{\mathrm{b}}(\mathbf{k}_X)$. The closed conic subset $\mathrm{SS}(F)$ of T^*X is defined by the following condition: $p \notin \mathrm{SS}(F)$ if and only if there exists an open neighborhood U of p such that for any $x \in \pi(U)$ and for any real-valued C^1-function φ defined on a neighborhood of x such that $\varphi(x) = 0$, $d\varphi(x) \in U$, one has

$$\mathrm{R}\Gamma_{\{\varphi \geqq 0\}}(F)_x = 0. \tag{3.1}$$

Kashiwara, Monteiro-Fernandes, and Schapira modified this definition and gave the notion of truncated microsupport.

Definition 3.2 ([KMS03a, Definition 3.4])**.** Let $F \in \mathsf{D}^{\mathrm{b}}(\mathbf{k}_X)$, and fix an integer $k \in \mathbf{Z}$. The *truncated microsupport* $\mathrm{SS}_k(F)$ is the closed conic subset of T^*X defined by the following condition: $p \notin \mathrm{SS}_k(F)$ if and only if there exists an open conic neighborhood U of p such that for any $x \in \pi(U)$ and for any real-valued C^1-function φ defined on a neighborhood of x such that $\varphi(x) = 0$, $d\varphi(x) \in U$, one has

$$H^j_{\{\varphi \geqq 0\}}(F)_x = 0 \quad \text{for any } j \leqq k. \tag{3.2}$$

For other equivalent conditions of the definition, see [KMS03a]. Truncated microsupports satisfy the following functorial property under proper direct images.

Proposition 3.3 ([KMS03a, Proposition 4.3])**.** Let $f\colon X \to Y$ be a morphism of manifolds and let $F \in \mathsf{D}^{\mathrm{b}}(\mathbf{k}_X)$ such that f is proper on $\mathrm{supp}(F)$. Then for any $k \in \mathbf{Z}$,

$$\mathrm{SS}_k(\mathrm{R}f_*F) \subset f_\pi f_d^{-1}(\mathrm{SS}_k(F)). \tag{3.3}$$

Moreover if f is a closed embedding, the inclusion above is an equality.

For other functorial properties, see [KMS03a, FM07].

4 Truncated Noncharacteristic Deformation Lemma

In this section, we prove a truncated version of the noncharacteristic deformation lemma. First we state the result. After giving several examples, we prove Theorem 4.2. In the course of the proof, we use the key lemma (Lemma 4.4). Since its proof is rather long, we postpone the proof to Section **??**.

Before we state the truncated version, let us review the original version of the noncharacteristic deformation lemma.

Proposition 4.1 ([KS90, Proposition 2.7.2])**.** Let X be a Hausdorff space. Let $F \in \mathsf{D}^+(\mathbf{k}_X)$, and let $(U_t)_{t\in\mathbf{R}}$ be a family of open subsets of X. We assume the following conditions.

(i) $U_t = \bigcup_{s<t} U_s$, for all $t \in \mathbf{R}$.

(ii) For all pairs (s,t) with $s \leqq t$, the set $\overline{U_t \setminus U_s} \cap \operatorname{supp} F$ is compact.

(iii) Setting $Z_s \coloneqq \bigcap_{t>s} \overline{U_t \setminus U_s}$, we have for all pairs (s,t) with $s \leqq t$ and all $x \in Z_s \setminus U_t$,
$$(\mathrm{R}\Gamma_{X\setminus U_t}(F))_x = 0.$$

Then, for all $t \in \mathbf{R}$, we have the isomorphisms
$$\mathrm{R}\Gamma\left(\bigcup_{s\in\mathbf{R}} U_s; F\right) \xrightarrow{\sim} \mathrm{R}\Gamma\left(U_t; F\right).$$

Our result is as follows.

Theorem 4.2 (Truncated Noncharacteristic Deformation Lemma)**.** Let X be a Hausdorff space. Let $F \in \mathsf{D}^+(\mathbf{k}_X)$, and let $(U_t)_{t\in\mathbf{R}}$ be a family of open subsets of X. Let $k \in \mathbf{Z}$. For $s \in \mathbf{R}$, set $Z_s \coloneqq \bigcap_{t>s} \overline{U_t \setminus U_s}$. We assume the following conditions.

(i) $U_t = \bigcup_{s<t} U_s$, for all $t \in \mathbf{R}$.

(ii) For all pairs (s,t) with $s \leqq t$, the set $\overline{U_t \setminus U_s} \cap \operatorname{supp} F$ is compact.

(iii) For all $s \in \mathbf{R}$ and all $x \in Z_s \setminus U_s$, we have
$$(H^j_{X\setminus U_s}(F))_x = 0 \quad \text{for all } j \leqq k.$$

(iv) For all pairs (s,t) with $s < t$, we have $Z_s \subset U_t$.

Then, for all $t \in \mathbf{R}$, we have the isomorphisms

$$H^j\left(\bigcup_{s\in\mathbf{R}} U_s; F\right) \xrightarrow{\sim} H^j\left(U_t; F\right) \quad \text{for all } j < k,$$

and the monomorphisms

$$H^k\left(\bigcup_{s\in\mathbf{R}} U_s; F\right) \hookrightarrow H^k\left(U_t; F\right).$$

Before we prove Theorem 4.2, let us give several examples of families of deformation.

Example 4.3. The condition (iv) of Theorem 4.2 is a new addition to the original version (Proposition 4.1). Therefore let us give an example of $(U_t)_{t\in\mathbf{R}}$ that does not satisfy the condition (iv).

(i) Let $X = \mathbf{R}^2$, and for a real number t define the function

$$\varphi_t(x) \coloneqq \begin{cases} \max\left(0, t(2-|x|)\right) & (|x| < 2), \\ 0 & (|x| \geqq 2). \end{cases}$$

Then let us define the open subset U_t of X by

$$U_t \coloneqq \left\{(x,y) \in X \,;\, y < \varphi_t(x)\right\}.$$

Then the family $(U_t)_t$ does not satisfy the condition (iv). Indeed, if $0 < s < t$, we have

$$Z_s = \left\{(x,y) \in X;\ y = \varphi_s(x), |x| \leqq 2\right\}$$

and the point $(2,0) \in Z_s$ is not contained in U_t. Moreover, let us define a locally closed subset S of X by

$$S \coloneqq \left\{(x,y) \in X;\ (x+2)^2 + (y-1)^2 = 1\right\} \setminus \left\{(-1,0)\right\}.$$

We denote by $\mathbf{k}_S$ the sheaf on X as in Section 2. If $j \leqq 0$, we have $H^j_{X\setminus U_t}(\mathbf{k}_S)_x = 0$ for all $t \geqq s$ and $x \in Z_s \setminus U_t$, but the morphism $H^0(U_1; \mathbf{k}_S) = \mathbf{k} \to 0 = H^0(U_0; \mathbf{k}_S)$ is not injective. This gives an example that satisfies the conditions (i)–(iii) but not (iv), which results in the consequence of the proposition does not hold. Hence the condition (iv) is necessary.

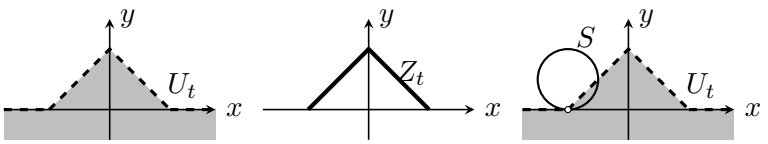


Figure 1: example (i)

Let us, in turn, give two examples of $(U_t)_{t\in\mathbf{R}}$ satisfying the condition (iv). In all the examples below, X is $\mathbf{R}^2$.

(ii) For a real number t, define the open set U_t of X by

$$U_t \coloneqq \begin{cases} \{(x,y) \in X;\ x^2+y^2 < t\} & (t > 0), \\ \varnothing & (t \leqq 0). \end{cases}$$

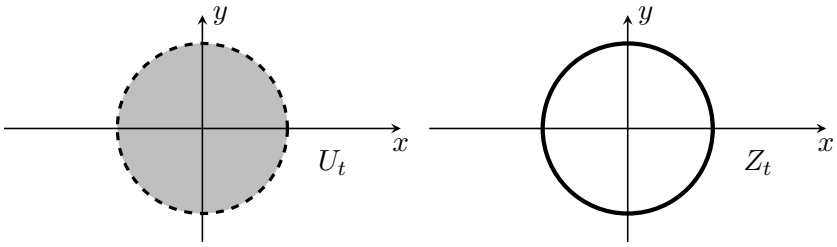


Figure 2: example (ii)

(iii) For a real number t, define the function $\varphi_t\colon \mathbf{R} \to \mathbf{R}$ as follows. If $t \leqq 0$, set $\varphi_t(x) \equiv 0$. If $t > 0$, set

$$\varphi_t(x) \coloneqq \begin{cases} \sqrt{t^2-x^2} & (|x| < t), \\ 0 & (|x| \geqq t). \end{cases}$$

Then define the open set U_t of X by

$$U_t \coloneqq \{(x,y) \in X;\ y < \varphi_t(x)\}\,.$$

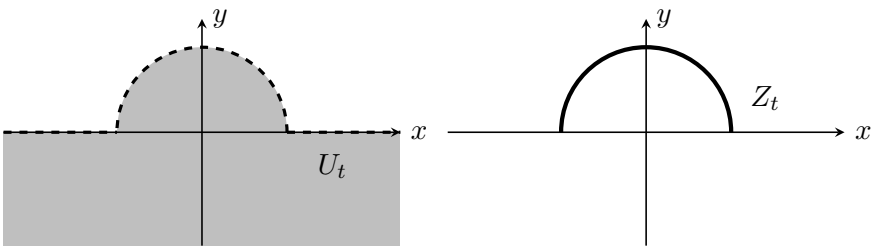


Figure 3: example (iii)

Proof of Theorem 4.2 Consider the following conditions for $j \in \mathbf{Z}$ and $s \in \mathbf{R}$:

$(a)^j_s$ $\qquad \mu^j_s \colon \varinjlim_{t>s} H^j(U_t; F) \xrightarrow{\sim} H^j(U_s; F),$

$(a')^k_s$ $\qquad \mu^k_s \colon \varinjlim_{t>s} H^k(U_t; F) \hookrightarrow H^k(U_s; F),$

$(b)^j_s$ $\qquad \lambda^j_s \colon H^j(U_s; F) \xrightarrow{\sim} \varprojlim_{r<s} H^j(U_r; F).$

Let us prove, for all $s \in \mathbf{R}$, $(a)^j_s$ for $j < k$ and $(a')^k_s$. We may assume that $\overline{U_t \setminus U_s}$ is compact for any $s \leqq t$ by replacing X by $\operatorname{supp} F$. Consider the distinguished triangle

$$\mathrm{R}\Gamma_{X \setminus U_s}(F) \to F \to \mathrm{R}\Gamma_{U_s}(F) \xrightarrow{+1}.$$

Let us endow $Z_s \cup U_s$ with the induced topology, and denote by $i_s \colon Z_s \cup U_s \hookrightarrow X$ the natural continuous map. Applying the functor

$$(\ \cdot\ )|_{Z_s \cup U_s} = i_s^{-1}$$

to the distinguished triangle above, we have

$$\mathrm{R}\Gamma_{X \setminus U_s}(F)|_{Z_s \cup U_s} \to F|_{Z_s \cup U_s} \to \mathrm{R}\Gamma_{U_s}(F)|_{Z_s \cup U_s} \xrightarrow{+1}.$$

Again applying to this triangle the functor $\mathrm{R}\Gamma(Z_s \cup U_s;\ \cdot\ )$, we have

(4.1)

$$\mathrm{R}\Gamma(Z_s \cup U_s; \mathrm{R}\Gamma_{X \setminus U_s}(F)|_{Z_s \cup U_s}) \to \mathrm{R}\Gamma(Z_s \cup U_s; F|_{Z_s \cup U_s}) \to \mathrm{R}\Gamma(U_s; F) \xrightarrow{+1}.$$

By the hypothesis (iii) and the definition of a local cohomology, we see that

$$H^j_{X \setminus U_s}(F)\Big|_{Z_s \cup U_s} = 0 \quad \text{for any } j \leqq k.$$

Therefore we have

$$H^j\left(Z_s \cup U_s; \mathrm{R}\Gamma_{X\setminus U_s}(F)|_{Z_s\cup U_s}\right) = 0 \quad \text{for any } j \leqq k.$$

Thus we have from (4.1)

$$H^j(Z_s \cup U_s; F|_{Z_s\cup U_s}) \xrightarrow{\sim} H^j(U_s; F) \quad \text{for any } j < k, \tag{4.2}$$

$$H^k(Z_s \cup U_s; F|_{Z_s\cup U_s}) \hookrightarrow H^k(U_s; F). \tag{4.3}$$

Therefore, if we show Lemma 4.4 below, whose proof will be given in Section **??**, we have from Lemma 4.4 and (4.2)

$$\begin{aligned}\varinjlim_{t>s} H^j(U_t; F) &\cong H^j(Z_s \cup U_s; F|_{Z_s\cup U_s}) \\ &\xrightarrow{\sim} H^j(U_s; F)\end{aligned}$$

for any $j < k$, and $(a)^j_s$ follows. Similarly, we have from Lemma 4.4 and (4.3)

$$\begin{aligned}\varinjlim_{t>s} H^k(U_t; F) &\cong H^k(Z_s \cup U_s; F|_{Z_s\cup U_s}) \\ &\hookrightarrow H^k(U_s; F)\end{aligned}$$

and $(a')^k_s$ follows.

Lemma 4.4. For any $j \in \mathbf{Z}$ and $s \in \mathbf{R}$, there exists a natural isomorphism

$$\varinjlim_{t>s} H^j(U_t; F) \cong H^j(Z_s \cup U_s; F|_{Z_s\cup U_s}).$$

Proof of Lemma 4.4 First we prove that for $F \in \mathrm{Mod}(\mathbf{k}_X)$, the natural morphism

$$\varinjlim_{t>s} \Gamma(U_t; F) \xrightarrow{\sim} \Gamma(Z_s \cup U_s; F) \tag{4.4}$$

is an isomorphism. Consider the following diagram.

$$\begin{array}{ccc}
\varinjlim_{t>s} \Gamma(U_t; F) & & \\
\downarrow{\scriptstyle \tilde{\phi}} & \searrow & \\
\varinjlim_{U\supset Z_s} \Gamma(U \cup U_s; F) & \xrightarrow[\phi]{} & \Gamma(Z_s \cup U_s; F|_{Z_s\cup U_s}) \\
\downarrow & & \downarrow \\
\varinjlim_{U\supset Z_s} \Gamma(U; F) & \xrightarrow[\psi]{\sim} & \Gamma(Z_s; F|_{Z_s}).
\end{array}$$

Let us begin by showing that the morphism

$$\tilde{\phi}\colon \varinjlim_{t>s} \Gamma(U_t; F) \to \varinjlim_{U \supset Z_s} \Gamma(U \cup U_s; F)$$

is an isomorphism. It is enough to show that $(U_t)_{t>s}$ is a cofinal family of $(U \cup U_s)_{U \supset Z_s}$. Let U be an open neighborhood of Z_s. Then we shall show by contradiction that there exists a real number $t > s$ such that $U_t \subset U \cup U_s$. Let us assume that $U_t - (U \cup U_s) \neq \varnothing$ for any $t > s$. Then define the family of subsets $(A_r)_{s<r<t}$ of $\overline{U_t - U_s}$ by $A_r \coloneqq U_r - (U \cup U_s)$. This $(A_r)_{s<r<t}$ satisfies the finite intersection property. Indeed, for finite real numbers $s < r_0 < \cdots < r_m < t$, it follows from the assumption for contradiction that

$$\begin{aligned} A_{r_0} \cap \cdots \cap A_{r_m} &= A_{r_0} \\ &= U_{r_0} - (U \cup U_s) \\ &\neq \varnothing. \end{aligned}$$

Therefore, since $\overline{U_t - U_s}$ is compact, we have $\bigcap_{s<r<t} \overline{A_r} \neq \varnothing$. However, we have from the hypothesis $Z_s \subset U$ that

$$\begin{aligned} \varnothing \neq \bigcap_{s<r<t} \overline{A_r} &= \bigcap_{s<r<t} \overline{U_r - (U \cup U_s)} \\ &= \bigcap_{s<r<t} (\overline{U_r - U_s} \cap \overline{U_r - U}) \\ &= \bigcap_{s<r<t} \overline{U_r - U_s} \cap \bigcap_{s<r<t} \overline{U_r - U} \\ &= Z_s \cap \bigcap_{s<r<t} \overline{U_r - U} \\ &= Z_s \cap \bigcap_{s<r<t} \overline{U_r} \cap \overline{X - U} \\ &= Z_s \cap \bigcap_{s<r<t} \overline{U_r} \cap (X - U) \\ &= Z_s \cap (X - U) \cap \bigcap_{s<r<t} \overline{U_r} \\ &= \varnothing. \end{aligned}$$

This is a contradiction, and it follows that $\tilde{\phi}$ is an isomorphism.

Since $\tilde{\phi}$ is an isomorphism, it is enough to show that

$$\phi\colon \varinjlim_{U \supset Z_s} \Gamma(U \cup U_s; F) \to \Gamma(Z_s \cup U_s; F|_{Z_s \cup U_s}) \tag{4.5}$$

is an isomorphism. Since X is Hausdorff and Z_s is compact, the natural morphism

$$\psi\colon \varinjlim_{U\supset Z_s} \Gamma(U;F) \to \Gamma(Z_s;\, F|_{Z_s\cup U_s})$$

is an isomorphism by Proposition 2.2. Since U_s is open, we have another isomorphism

$$\psi'\colon \varinjlim_{V\supset U_s} \Gamma(V;F) \to \Gamma(U_s;F).$$

The injectivity of ϕ follows from the definition of a sheaf and the injectivities of ψ and ψ'.

Let us prove that ϕ is surjective. Let $\sigma \in \Gamma(Z_s\cup U_s;F)$ be a section of $F|_{Z_s\cup U_s}$. By the isomorphism ψ above, we can extend $\sigma|_{Z_s}$ to an open neighborhood U of Z_s in X. More precisely, there exists a section τ on U such that $\tau|_{Z_s} = \sigma|_{Z_s}$. On the other hand, it follows from the compatibility of restrictions of a section that $\tau|_{Z_s} = (\tau|_{(Z_s\cup U_s)\cap U})\Big|_{Z_s}$. By the definition of a section on the closed subset Z_s of $(Z_s\cup U_s)\cap U$, we can find an open subset W of $(Z_s\cup U_s)\cap U$ such that

$$\tau|_W = \sigma|_W.$$

By the definition of the induced topology of $(Z_s\cup U_s)\cap U$, there exists an open set $\widetilde{W}$ of X such that

$$W = ((Z_s\cup U_s)\cap U)\cap \widetilde{W} \quad \text{and} \quad \widetilde{W}\subset U.$$

Thus we have

$$U_s\cap W = U_s\cap \widetilde{W}, \quad \sigma|_{U_s\cap\widetilde{W}} = \tau|_{U_s\cap\widetilde{W}}. \tag{4.6}$$

For this $\widetilde{W}$, the pair $\left(U_s,\widetilde{W}\right)$ is an open covering of $\widetilde{W}\cup U_s$. Thus by (4.6), we can glue $\sigma|_{U_s}$ and $\tau|_{\widetilde{W}}$ together and get a section $\widetilde{\sigma}$ such that

$$\widetilde{\sigma}|_{Z_s\cup U_s} = \sigma, \quad \widetilde{\sigma}|_{\widetilde{W}} = \tau|_{\widetilde{W}}.$$

This proves the surjectivity of ϕ.

Next let $F\in \mathsf{D}^+(\mathbf{k}_X)$. Then it follows from (4.4) and Lemma A.1 below that

$$\varinjlim_{t>s} H^j(U_t;F) \cong H^j(Z_s\cup U_s;\, F|_{Z_s\cup U_s}).$$

This completes the proof. □

Let us come back to the proof of Theorem 4.2. We prove $(b)^j_s$ for all $j \in \mathbf{Z}$ and $s \in \mathbf{R}$ by induction on j. Since $F \in \mathsf{D}^+(\mathbf{k}_X)$, we have $(b)^j_s$ for $j \ll 0$ and $s \in \mathbf{R}$. Hence there exists an integer $j_0 < k$ such that $(b)^j_s$ holds for any $j < j_0$ and $s \in \mathbf{R}$. Then assume $(b)^j_s$ is proved for all $j < j_0$ and $s \in \mathbf{R}$. Since we have $(a)^j_s$ and $(b)^j_s$ for each $j < j_0$ and each $s \in \mathbf{R}$, we can apply Kashiwara's lemma (Proposition 2.4) and have

$$H^j(U_t; F) \xrightarrow{\sim} H^j(U_s; F) \quad \text{for } j < j_0 \text{ and } s \leqq t.$$

Fix $s \in \mathbf{R}$ and consider the projective system $\left(H^{j_0-1}(U_{s-1/n}); F\right)_{n\in\mathbf{N}}$. This system satisfies the ML condition. Therefore, by applying Proposition 2.5 (iii), we have

$$H^j(U_s; F) \xrightarrow{\sim} \varprojlim_n H^j\left(U_{s-\frac{1}{n}}; F\right).$$

Since $(s-1/n)_n$ is a cofinal family of $(t)_{s>t}$, we have $(b)^{j_0}_s$. Then by induction on j, we have $(b)^j_s$ for $j < k$ and $s \in \mathbf{R}$. Again by applying Proposition 2.5 to the projective system $\left(H^j(U_n; F)\right)_n$, we obtain

$$\begin{aligned} H^j\left(\bigcup_{s\in\mathbf{R}} U_s; F\right) &\cong H^j\left(\bigcup_{n\in\mathbf{N}} U_n; F\right) \\ &\xrightarrow{\sim} \varprojlim_n H^j\left(U_n; F\right) \end{aligned}$$

and for each n, we have by Kashiwara's lemma (Proposition 2.4) that

$$H^j\left(U_n; F\right) \cong H^j(U_s; F)$$

for $n \geqq s$.

Finally we replace $(a)^j_s$ by $(a')^k_s$ for $j = k$, and obtain

$$H^k\left(\bigcup_{s\in\mathbf{R}} U_s; F\right) \hookrightarrow H^j\left(U_s; F\right).$$

This completes the proof. □

5 Truncated Microlocal Morse theory

In this section, we show a truncated version of the relative microlocal Morse lemma as an application of the truncated noncharacteristic deformation

lemma. After that, considering the absolute version, we show a more familier statement as the microlocal Morse lemma but, of corse, in a truncated setting. We can consider the last proposition as a global version of famous Holmgren's uniqueness theorem. In order to show it, we give an example at the end of this section.

Let $f\colon Y \to X$ be a morphism of manifolds, and φ a real valued function of class C^1 on Y. For a real number $t \in \mathbf{R}$, we set

$$\begin{aligned} Y_t &\coloneqq \{y \in Y\,;\, \varphi(y) < t\} = \varphi^{-1}(]-\infty, t[), \\ \widetilde{Y}_t &\coloneqq \{y \in Y\,;\, \varphi(y) \leqq t\} = \varphi^{-1}(]-\infty, t]). \end{aligned}$$

Let us define the following morphisms:

$$\begin{array}{ccc} Y_t & \xrightarrow{j_t} & Y \\ & \searrow^{f_t} & \downarrow f \\ & & X, \end{array} \qquad \begin{array}{ccc} \widetilde{Y}_t & \xrightarrow{\tilde{j}_t} & Y \\ & \searrow^{f_t} & \downarrow f \\ & & X. \end{array}$$

Proposition 5.1. Let $G \in \mathsf{D}^{\mathrm{b}}(\mathbf{k}_Y)$, and assume $f_t\colon \operatorname{supp}(G) \cap \overline{Y_t} \to X$ is proper for all $t \in \mathbf{R}$. Let k be an integer and $t_\circ$ a real number. Assume that, for any $y \in Y \setminus \widetilde{Y}_{t_\circ}$, we have

$$d\varphi(y) \notin \left(\mathrm{SS}_k(G) + f_d\left(Y \underset{X}{\times} T^*X\right)\right). \tag{5.1}$$

Then for $t > t_\circ$, the natural morphism

$$\mathrm{R}^j f_* G \to \mathrm{R}^j f_{t*} j_t^{-1} G$$

is an isomorphism for $j < k$ and a monomorphism for $j = k$.

Proof. Following [KS90], let $\tilde{f}$ be the morphism $(f, \varphi)\colon Y \to X \times \mathbf{R}$, and q and $\tilde{\varphi}$ be the first and second projections defined on $X \times \mathbf{R}$ respectively. Then $\tilde{f}$ is proper on $\operatorname{supp}(G)$ and $f = q \circ \tilde{f}$. Let i_t be the open embedding $X \times]-\infty, t[\hookrightarrow X \times \mathbf{R}$ and $\tilde{i}_t$ the closed embedding $X \times]-\infty, t] \hookrightarrow X \times \mathbf{R}$. Set $q_t = q \circ i_t$ and $\tilde{q}_t = q \circ \tilde{i}_t$. These morphisms can be visualized by the following diagrams, in both of which the left squares are cartesian.

$$\begin{array}{ccccc} Y & \xrightarrow{\tilde{f}} & X \times \mathbf{R} & \xrightarrow{q} & X\,, \\ {\scriptstyle j_t}\uparrow & \square & \uparrow{\scriptstyle i_t} & \nearrow_{q_t} & \\ Y_t & \xrightarrow{\hat{f}_t} & X \times]-\infty, t[& & \end{array} \qquad \begin{array}{ccccc} Y & \xrightarrow{\tilde{f}} & X \times \mathbf{R} & \xrightarrow{q} & X\,. \\ {\scriptstyle \tilde{j}_t}\uparrow & \square & \uparrow{\scriptstyle \tilde{i}_t} & \nearrow_{\tilde{q}_t} & \\ \widetilde{Y}_t & \xrightarrow{\check{f}_t} & X \times]-\infty, t] & & \end{array}$$

Then, setting $\widetilde{G} \coloneqq \mathrm{R}\tilde{f}_* G$, we have $\mathrm{R}f_* G \cong \mathrm{R}q_* \widetilde{G}$ and

$$\mathrm{R}f_{t*} j_t^{-1} G \cong \mathrm{R}q_{t*} i_t^{-1} \widetilde{G}, \qquad \mathrm{R}\tilde{f}_{t*} \tilde{j}_t^{-1} G \cong \mathrm{R}\tilde{q}_{t*} \tilde{i}_t^{-1} \widetilde{G},$$
$$\mathrm{R}f_{t!} j_t^{-1} G \cong \mathrm{R}q_{t!} i_t^{-1} \widetilde{G}, \qquad \mathrm{R}\tilde{f}_{t*} \tilde{j}_t^{!} G \cong \mathrm{R}\tilde{q}_{t*} \tilde{i}_t^{!} \widetilde{G}.$$

Therefore, it is enough to prove the result for $\widetilde{G}$, q, i_t, $\tilde{\varphi}$, instead of G, f, j_t, φ.

It follows from the hypothesis (5.1) that

$$\mathrm{SS}_k(\widetilde{G}) \cap (T^*X \times \{(t;\tau); t > t_\circ\}) \subset T^*X \times \{(t;\tau); \tau \leqq 0\}\,. \tag{5.2}$$

Indeed, by hypothesis, for any $(y;\eta) \in \mathrm{SS}_k(G)$ and $(f(x);\xi) \in Y \times_X T^*X$, we have

$$d\varphi(y) \neq \eta + f^*\xi. \tag{5.3}$$

If $(x,t;\xi,\tau)$ is a point of $\mathrm{SS}_k(\widetilde{G}) \cap (T^*X \times \{(t;\tau); t > t_\circ\})$, then, by the formula (3.3), there exists $(y;\eta) \in \mathrm{SS}_k(G)$ such that

$$\tilde{f}(y) = (x,t) \text{ and } \eta = \tilde{f}^*(\xi,\tau).$$

For this η, we have

$$\eta = \tilde{f}^*(\xi,\tau) = f^*\xi + \varphi^*\tau = f^*\xi + \tau d\varphi(y).$$

If $\tau > 0$, we get

$$d\varphi(y) = \frac{1}{\tau}\eta - \frac{1}{\tau} f^*\xi$$

and this contradicts (5.3) because $\mathrm{SS}_k(G)$ is conic.

We shall show that, for all $x \in X$, the morphism

$$\left(\mathrm{R}^j q_* \widetilde{G}\right)_x \to \left(\mathrm{R}^j q_{t*} i_t^{-1} \widetilde{G}\right)_x$$

is an isomorphism if $j < k$ and a monomorphism if $j = k$. By definition, it suffices to show the same claims for the morphism

$$\varinjlim_{x \in U} H^j\left(U \times \mathbf{R}; \widetilde{G}\right) \to \varinjlim_{x \in U} H^j\left(U \times]-\infty, t[\,; \widetilde{G}\right).$$

Let U be an open neighborhood of x in X contained in a local chart, and diffeomorphic to the unit open ball, centered at 0. Such open sets form a

fundamental system of neighborhoods of x. Thus we show the result for the morphism

$$H^j\left(U \times \mathbf{R}; \widetilde{G}\right) \to H^j\left(U \times]-\infty, t[\,; \widetilde{G}\right)$$

when U is in the form as above.

Fix $t > t_\circ$, and identify U with the open unit ball in Euclidean space using the chosen coordinates. We construct a deformation family $(U_s)_{s\in\mathbf{R}}$ whose initial member is $U\times]-\infty, t[$. Choose a smooth function $\rho\colon\ [0, +\infty[\to [0, 1]$ satisfying

$$\begin{aligned} \rho(u) &= 1 && (0 \leqq u \leqq 1/2), \\ \rho'(u) &< 0 && (1/2 < u < 1), \\ \rho(u) &= 0 && (u \geqq 1). \end{aligned}$$

For $s \in \mathbf{R}$, set

$$h_s(x) \coloneqq \begin{cases} t + (s-1)\rho\left(\dfrac{s+1}{s}|x|\right), & s > 1, \\ t, & s \leqq 1, \end{cases} \qquad U_s \coloneqq \{(x, x') \in U\times\mathbf{R}\ ;\ x' < h_s(x)\}.$$

Each h_s is smooth, since ρ is constant near 0. In particular,

$$U_1 = U \times]-\infty, t[\,, \qquad \bigcup_{s\in\mathbf{R}} U_s = U \times \mathbf{R}.$$

The second equality follows because $h_s(x) \to +\infty$ for every $x \in U$ as $s \to +\infty$.

We verify conditions (i)–(iv) of Theorem 4.2, with ambient space $U \times \mathbf{R}$. For each x, the function $s \mapsto h_s(x)$ is continuous and nondecreasing. Hence $U_s = \bigcup_{r<s} U_r$, proving condition (i). For $s < r$ with $r > 1$,

$$\overline{U_r \setminus U_s} \subset \left\{x\ ;\ |x| \leqq \frac{r}{r+1}\right\} \times [t, t + r - 1],$$

and the set on the right is compact in $U \times \mathbf{R}$. If $r \leqq 1$, then $U_r = U_s$. This proves condition (ii).

Set $Z_s = \bigcap_{r>s} \overline{U_r \setminus U_s}$, where closures are taken in $U \times \mathbf{R}$. For $s < 1$, we have $Z_s = \varnothing$. For $s \geqq 1$, continuity in s and the preceding support bound give

$$Z_s \subset \left\{(x, h_s(x))\ ;\ |x| \leqq \frac{s}{s+1}\right\}.$$

If $r > s \geqq 1$ and $|x| \leqq s/(s+1)$, then $\rho((r+1)|x|/r) > 0$ and $h_r(x) > h_s(x)$. Therefore $Z_s \subset U_r$ for every $r > s$, proving condition (iv).

Finally, near a point $z = (x, h_s(x)) \in \partial U_s$, the complement of U_s is defined by $g_s \geqq 0$, where

$$g_s(x, x') = x' - h_s(x), \qquad dg_s(z) = (-dh_s(x), 1).$$

Since $h_s(x) \geqq t > t_\circ$, estimate (5.2) implies $dg_s(z) \notin \mathrm{SS}_k(\widetilde{G})$. By Definition 3.2,

$$\big(H^j_{(U\times\mathbf{R})\setminus U_s}(\widetilde{G})\big)_z = 0 \qquad (j \leqq k).$$

Here $\widetilde{G}$ is restricted to $U \times \mathbf{R}$. Since $Z_s \subset \partial U_s$, condition (iii) follows.

Applying Theorem 4.2 at the parameter value $s = 1$, we obtain the restriction morphism

$$\begin{aligned} H^j(U \times \mathbf{R}; \widetilde{G}) &= H^j\left(\bigcup_{s\in\mathbf{R}} U_s; \widetilde{G}\right) \\ &\longrightarrow H^j(U_1; \widetilde{G}) \\ &= H^j\left(U \times]-\infty, t[\,; \widetilde{G}\right). \end{aligned}$$

It is an isomorphism for $j < k$ and a monomorphism for $j = k$, as required. □

Considering the case $X = \{\mathrm{pt}\}$, one has the absolute version of the microlocal Morse lemma.

Theorem 5.2. Let $F \in \mathsf{D}^{\mathrm{b}}(\mathbf{k}_M)$, and $\psi\colon M \to \mathbf{R}$ be a real-valued function of class C^1, proper on $\mathrm{supp}(F)$. Let $a < b$ be real numbers, and $k \in \mathbf{Z}$. We assume $d\psi(x) \notin \mathrm{SS}_k(F)$ for $a \leqq \psi(x) < b$. For $t \in \mathbf{R}$, we set $M_t \coloneqq \psi^{-1}(]-\infty, t[)$. Then, the restriction morphism

$$H^j(M_b; F) \to H^j(M_a; F)$$

is an isomorphism for each $j < k$ and a monomorphism for $j = k$.

Example 5.3. Let M be a real analytic manifold and X be its complexification. Let P be a linear partial differential operator in the analytic coefficient defined locally on X. By $\mathcal{M} \coloneqq \mathcal{D}_X/\mathcal{D}_X P$, we denote the coherent $\mathcal{D}_X$-module associated with P. For the theory of $\mathcal{D}_X$-modules, one can consult [Kas03]. Set $F \coloneqq \mathrm{R}\mathscr{H}om_{\mathcal{D}_M}(\mathcal{M}, \mathcal{B}_M)$. Let φ as Theorem 5.2. By

Holmgren's uniqueness theorem (cf. [Hor83, Theorem 8.6.5]), one can show the following estimate (see also [KMS03a, Example 3.8 (iv)]):

$$\mathrm{SS}_0(F) \subset \mathrm{Char}(\mathcal{M}) \cap T^*M.$$

By Theorem 5.2, it follows that the restriction morphism

$$H^0(M_b; F) \to H^0(M_a; F).$$

is injective if all the level hypersurfaces $\varphi^{-1}(t)$ $(a \leqq t < b)$ are non-characteristic for $\mathcal{M}$.

A An Isomorphism of Global Sections in Particular Cases

In this subsection, we shall show an isomorphism of global sections which is needed in the proof of Theorem 4.2.

Let X be a topological space, and S a subset of X. we denote the derived functor of the left exact functor $F \mapsto \Gamma(S; F|_S)$ by $\mathrm{R}\Gamma(S; \cdot\ )$. There exists a morphism of functors

$$\mathrm{R}\Gamma(S; F) \longrightarrow \mathrm{R}\Gamma(S; F|_S),$$

which is not an isomorphism in general. Suppose the set S has the following property: for any sheaf F, the canonical morphism

$$\varinjlim_{S \subset W} \Gamma(W; F) \longrightarrow \Gamma(S; F|_S)$$

is an isomorphism, where W runs through the open neighborhoods of S.

Note that in this case, $\mathrm{R}\Gamma(S; \cdot\ )$ can be computed by a flabby resolution. That is, if I is a flabby resolution of F, then

$$\mathrm{R}\Gamma(S; F) \simeq \Gamma(S; I|_S).$$

The right-hand side is generally not equal to $\mathrm{R}\Gamma(S; F|_S)$, because $I|_S$ is not necessarily a complex consisting of flabby sheaves on the space S. However, in a special case, the following theorem holds.

Lemma A.1. Let X be a Hausdorff topological space, $Z \subset X$ a compact set, and $U \subset X$ an open set, and assume $U \cap Z = \varnothing$. For $F \in \mathsf{D}^+(\mathbf{k}_X)$,

$$\mathrm{R}\Gamma(U \cup Z; F) \xrightarrow{\sim} \mathrm{R}\Gamma(U \cup Z; F|_{U \cup Z}).$$

In particular,

$$H^k(U \cup Z; F) \simeq \varinjlim_{Z \subset W} H^k(U \cup W; F),$$

where W runs through the open neighborhoods of Z.

Proof. First, if F is a sheaf, by the completely same argument in the proof of (4.5), we can show the isomorphism

$$\Gamma(U \cup Z; F|_{U \cup Z}) \simeq \varinjlim_{Z \subset W} \Gamma(U \cup W; F).$$

Let F be a sheaf and W an open neighborhood of Z. There exists an exact sequence

$$0 \to \Gamma(U \cup W; F) \to \Gamma(U; F) \oplus \Gamma(W; F) \to \Gamma(U \cap W; F) \simeq \Gamma(W; \Gamma_U F).$$

Furthermore, if F is flabby, the last morphism is surjective. Then, by letting W run through the neighborhoods of Z, we obtain an exact sequence

$$0 \to \Gamma(U \cup Z; F|_{U \cup Z}) \to \Gamma(U; F) \oplus \Gamma(Z; F|_Z) \to \Gamma(Z; (\Gamma_U F)|_Z),$$

and moreover, if F is flabby, the last morphism is surjective thanks to the exactness of the inductive limit.

Therefore, we have a distinguished triangle of derived functors

$$\mathrm{R}\Gamma(U \cup Z; F) \to \mathrm{R}\Gamma(U; F) \oplus \mathrm{R}\Gamma(Z; F) \to \mathrm{R}\Gamma(Z; \mathrm{R}\Gamma_U F) \xrightarrow{+1} . \tag{A.1}$$

Indeed, if F is flabby, so is $\Gamma_U F$, which implies

$$\mathrm{R}(\Gamma(Z; (\Gamma_U(\,\cdot\,))|_Z)) \simeq \mathrm{R}\Gamma(Z; \mathrm{R}\Gamma_U(\,\cdot\,)).$$

For simplicity, let $G = F|_{U \cup Z}$ and consider the triangle obtained by applying (A.1) assuming the whole space is $U \cup Z$. Since Z is compact and U is an open set in $U \cup Z$, we can use (A.1) to get

$$\mathrm{R}\Gamma(U \cup Z; G) \to \mathrm{R}\Gamma(U; G) \oplus \mathrm{R}\Gamma(Z; G) \to \mathrm{R}\Gamma(Z; \mathrm{R}\Gamma_U G) \xrightarrow{+1} .$$

Since Z is compact in $U \cup Z$, we have

$$\mathrm{R}\Gamma(Z; \mathrm{R}\Gamma_U G) \simeq \mathrm{R}\Gamma(Z; (\mathrm{R}\Gamma_U G)|_Z).$$

On the other hand, denoting the inclusions by $j \colon U \hookrightarrow U \cup Z$ and $\tilde{j} \colon U \hookrightarrow X$, we have

$$(\mathrm{R}\Gamma_U G)|_Z \simeq (\mathrm{R}j_*(F|_U))|_Z \simeq (\mathrm{R}\tilde{j}_*(F|_U))\big|_Z \simeq (\mathrm{R}\Gamma_U F)|_Z$$

and obtain

$$\mathrm{R}\Gamma(Z;\mathrm{R}\Gamma_U\, G) \simeq \mathrm{R}\Gamma(Z;\, (\mathrm{R}\Gamma_U\, F)|_Z) \simeq \mathrm{R}\Gamma(Z;\mathrm{R}\Gamma_U\, F).$$

The last isomorphism uses the fact that Z is compact in X. Similarly,

$$\mathrm{R}\Gamma(U;G) \simeq \mathrm{R}\Gamma(U;F), \quad \mathrm{R}\Gamma(Z;G) \simeq \mathrm{R}\Gamma(Z;F).$$

From the above, we obtain the triangle

$$\mathrm{R}\Gamma(U\cup Z;G) \to \mathrm{R}\Gamma(U;F) \oplus \mathrm{R}\Gamma(Z;F) \to \mathrm{R}\Gamma(Z;\mathrm{R}\Gamma_U\, F) \xrightarrow{+1},$$

and by comparing it with (A.1), we find that

$$\mathrm{R}\Gamma(U\cup Z;\, F|_{U\cup Z}) = \mathrm{R}\Gamma(U\cup Z;G) \simeq \mathrm{R}\Gamma(U\cup Z;F)$$

holds. □

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